

Spring 2024/2025

Class: Human Cytogenetics (BT 456)

Time: Mon, Wed: 10:30-11:30

Dr. Asem Alkhateeb

Place: Determined by registration

Office hours: to be determined

Office: PH4-L0

Topic	Chapter	Weeks
Elements of medical cytogenetics	1	1
Chromosome analysis	2	2
Origins and consequences of chromosome pathology	3	3,4
Autosomal reciprocal translocations	5	5
Sex chromosome translocations	6	6
Robertsonian translocations	7	7
Insertions	8	7
Inversions	9	8
Down syndrome, aneuploidy, polyploidy	13	9
Autosomal deletions & duplications	14	10
Uniparental disomy and disorders of imprinting	18	11
Reproductive failure	19	12
Prenatal testing procedures	20	13
Chromosome abnormalities detected at prenatal diagnosis	21	14
Preimplantation genetic diagnosis	22	15

Grade Distribution:

First exam	20%	
Second exam	20%	
Lab grades	30%	(includes quizzes, reports, mid exam, and final)
Final exam	30%	
Total	100%	

Each exam will have ~10% bonus questions. This bonus will not be present in make-up exams.

Textbook:

Chromosome Abnormalities and Genetic Counseling. Gardner and Amor. 5th ed. **2018**. [Available at library reserve shelf]

Attendance: You are required to attend all classes. Missing 20% of class will result in dismissal from class. No excuses are accepted beyond 20% limit according to University Regulations.

Course description: The major goal of this course will be to introduce all students to the basic principles of cytogenetics as they apply to basic and clinical genetics. The focus of the course will include chromosome structure and function, the underlying features of cytogenetic disorders, as well as the common numerical and structural abnormalities involved in the diagnosis and evaluation of patients with a likely cytogenetic disorder.

Course objectives:

- Describe the structure of the chromosome, karyotyping, banding patterns, other cytogenetic tools, and consequences of chromosome pathology
- Understand principles and consequences of chromosome translocations, insertions, and inversions
- Explain mechanisms and effects of aneuploidy, polyploidy, deletions, and duplications of the chromosomal material on human health
- Obtain hands-on experience with chromosome analysis techniques in Laboratory